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Detection of Drug Resistance Mutations in the Reverse Transcriptase Gene of HIV-1-Infected North Indian Population Failing First-Line Antiretroviral Therapy “A Follow-Up Cohort Study”

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Abstract

We aim to characterize the drug resistance mutations in reverse transcriptase gene of HIV-1 subtype C-infected North Indian population in those who are failing first-line antiretroviral therapy (ART) and if these mutations are associated with mortality. We also attempted the assessment of switch over to second-line antiretroviral therapy in these patients. Based on the immunological marker CD4 count (<350 cubic/mm), 192 HIV/AIDS patients were selected and viral load was estimated in those who were enrolled from December 2009 to November 2016. Based on viral load, genotyping was carried out in 57 HIV-1 isolates ($VL \geq 1,000$ copies/mL) by sequencing and drug resistance mutations were examined through the Stanford HIV Drug Resistance Database, USA. Among them, 21 (36.84%) first-line ART failure patients were shifted to second-line ART. These patients were followed for a period wide ranging from 10 months to 11 years. Drug resistance mutation M184V (ATG to GTA) (63.15%) associated with lamivudine and abacavir and K103N (AAG or AAA to AAU) (36.84%) associated with efavirenz and nevirapine were predominantly identified in first-line ART failure patients. During follow-up, it was observed that 3 out of 21 who were in second-line ART died, whereas 9 out of 36 died who were in the first-line ART. No mutation could be associated with mortality although TAM-2 mutations were absent in patients who died. This study indorses the need for a facility for viral load estimation and resistance monitoring in each treatment failure patient and availability of appropriate antiretroviral therapies.

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